

考研竞赛凯哥 第8次周考

考试时间: 5月10日上午 08:30~10:00

讲解时间: 5月10日上午 10:00~12:00

注1: 本次周考的范围为《高等数学上册》《多元微分的概念与计算》

注2: 周考, 每周一次, 比较频繁, 所以题量减半, 考试时间减半, 且选填题与解答题的比例随机;

月考, 每月一次, 完全按照考研的标准进行设置.

注3: 无论周考还是月考, 目的都是查漏补缺和巩固所学知识, 所以题目的难度至少持平或略高于真题的平均难度, 大家一定不要因为模考成绩不理想而灰心, 考前犯的任何错误都应该珍惜!

1. 设 $f(x)$ 三阶可导, $g(x) = \begin{cases} \frac{1}{x^2} [e^{2x} - f(x)], & x \neq 0 \\ 1, & x = 0 \end{cases}$, 若 $g'(0) = 1$, 则 ()

A. $f(0) = 2$ B. $f'(0) = 1$ C. $f''(0) = 1$ D. $f'''(0) = 2$

解: $g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\frac{1}{x^2} [e^{2x} - f(x)] - 1}{x - 0} = \lim_{x \rightarrow 0} \frac{[e^{2x} - f(x)] - x^2}{x^3} = 1,$

$$\lim_{x \rightarrow 0} \frac{\left[1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + o(x^3)\right] - \left[f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + o(x^3)\right] - x^2}{x^3} = 1,$$

$$\text{故} \begin{cases} 1 - f(0) = 0 \\ 2 - f'(0) = 0 \\ \frac{2^2}{2} - \frac{f''(0)}{2} - 1 = 0 \\ \frac{2^3}{3!} - \frac{f'''(0)}{3!} = 1 \end{cases}, \text{解得} \begin{cases} f(0) = 1 \\ f'(0) = 2 \\ f''(0) = 2 \\ f'''(0) = 2 \end{cases}, \text{故选 D.}$$

2. $\int \frac{1}{1 + e^{\frac{x}{2}} + e^{\frac{x}{3}} + e^{\frac{x}{6}}} dx.$

解: 令 $e^{\frac{x}{6}} = t$, 故 $e^{\frac{x}{3}} = t^2$, $e^{\frac{x}{2}} = t^3$, $x = 6 \ln t$, $dx = \frac{6}{t} dt$, 代入得

$$I = \int \frac{1}{1 + t + t^2 + t^3} \frac{6}{t} dt = 6 \int \frac{1}{t(t+1)(t^2+1)} dt.$$

设 $\frac{1}{t(t+1)(t^2+1)} = \frac{A}{t} + \frac{B}{t+1} + \frac{Ct+D}{t^2+1}$, 解得 (过程略) $A=1, B=C=D=-\frac{1}{2}$.

$$\text{故 } I = 6 \int \left(\frac{1}{t} + \frac{-\frac{1}{2}}{t+1} + \frac{-\frac{1}{2}t + (-\frac{1}{2})}{t^2+1} \right) dt = 6 \ln t - 3 \ln(1+t) - \frac{3}{2} \ln(1+t^2) - 3 \arctan t + C'$$

$$= 6 \ln e^{\frac{x}{6}} - 3 \ln \left(1 + e^{\frac{x}{6}} \right) - \frac{3}{2} \ln \left(1 + e^{\frac{x}{3}} \right) - 3 \arctan \left(e^{\frac{x}{6}} \right) + C'.$$

3. 设函数 $z = z(x, y)$ 由方程 $F\left(x + \frac{z}{y}, y + \frac{z}{x}\right) = 0$ 确定, F 可微, 则 $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + xy =$ _____.

解: 令 $G(x, y, z) = F\left(x + \frac{z}{y}, y + \frac{z}{x}\right)$, 由隐函数求导法则——

$$\begin{aligned} & x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + xy \\ &= x \left(-\frac{G'_x}{G'_z} \right) + y \left(-\frac{G'_y}{G'_z} \right) + xy \\ &= (-x) \frac{F'_1 - \frac{z}{x^2} F'_2}{\frac{1}{y} F'_1 + \frac{1}{x} F'_2} + (-y) \frac{-\frac{z}{y^2} F'_1 + F'_2}{\frac{1}{y} F'_1 + \frac{1}{x} F'_2} + xy \\ &= \frac{-x^2 y F'_1 + y z F'_2}{x F'_1 + y F'_2} + \frac{x z F'_1 - x y^2 F'_2}{x F'_1 + y F'_2} + \frac{x^2 y F'_1 + x y^2 F'_2}{x F'_1 + y F'_2} \\ &= \frac{y z F'_2 + x z F'_1}{x F'_1 + y F'_2} \\ &= \frac{z (y F'_2 + x F'_1)}{x F'_1 + y F'_2} \\ &= z. \end{aligned}$$

4. $\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n \frac{k}{n+k} \right)^{\frac{3}{e^n-1}}.$

解:

$$\begin{aligned} I &= e^{\lim_{n \rightarrow \infty} \left(e^{\frac{3}{e^n-1}} \right) \cdot \ln \prod_{k=1}^n \frac{k}{n+k}} \\ &= e^{3 \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{k=1}^n \ln \frac{\frac{k}{n}}{1 + \frac{k}{n}}} \\ &= e^{3 \int_0^1 [\ln x - \ln(1+x)] dx} \\ &= e^{3 \left[x \ln x \Big|_0^1 - \int_0^1 x \cdot \frac{1}{x} dx - x \ln(1+x) \Big|_0^1 + \int_0^1 \frac{x+1-1}{1+x} dx \right]} \\ &= e^{3[0-1-\ln 2+1-\ln 2]} \\ &= e^{-6 \ln 2} = e^{\ln 2^{-6}} = \frac{1}{2^6} = \frac{1}{64} \end{aligned}$$

5. 设 $f(x, y) = \begin{cases} (xy + a|x| + b\sqrt{|y|}) \arctan \frac{1}{|x| + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$, 则下列正确的是 ()

- A. $f(x, y)$ 在 $(0, 0)$ 处的连续性与 a, b 的取值有关
- B. $f(x, y)$ 在 $(0, 0)$ 处偏导存在的充要条件是 $ab = 0$
- C. $f(x, y)$ 在 $(0, 0)$ 处可微的充要条件是 $f(x, y)$ 在 $(0, 0)$ 处偏导存在
- D. 若 $f(x, y)$ 在 $(0, 0)$ 处可微, 则 $(0, 0)$ 是 $f(x, y)$ 的极值点

解: A 选项: 一定连续, 与 a, b 无关;

$$B \text{ 选项: } f'_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x-0} = \lim_{x \rightarrow 0} \frac{a \cdot |x| \cdot \frac{\pi}{2}}{x} = a \cdot \frac{\pi}{2} \cdot \lim_{x \rightarrow 0} \frac{|x|}{x} \text{ 要存在 } \iff a=0,$$

$$\text{同理 } f'_y(0,0) = \lim_{y \rightarrow 0} \frac{f(0,y) - f(0,0)}{y-0} = \lim_{y \rightarrow 0} \frac{b \cdot \sqrt{|y|} \cdot \frac{\pi}{2}}{y} = \text{要存在 } \iff b=0,$$

综上, $f(x,y)$ 在 $(0,0)$ 可偏导 $\iff a=0$ 且 $b=0$;

C 选项: $f(x,y)$ 在 $(0,0)$ 可偏导 $\iff a=0$ 且 $b=0$, 此时

$$\begin{aligned} & \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{[f(x,y) - f(0,0)] - 0 \cdot (x-0) - 0 \cdot (y-0)}{\sqrt{x^2 + y^2}} \\ &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{\sqrt{x^2 + y^2}} \arctan x \cdot \frac{1}{|x| + y^2} = \frac{\pi}{2} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{\sqrt{x^2 + y^2}} \end{aligned}$$

$$\text{其中 } 0 \leq \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| = \frac{|x| |y|}{\sqrt{x^2 + y^2}} \leq \frac{\frac{1}{2}(x^2 + y^2)}{\sqrt{x^2 + y^2}} = \frac{1}{2} \sqrt{x^2 + y^2} \rightarrow 0$$

$$\text{由夹逼: } \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| = 0 \Rightarrow \text{可微!}$$

D 选项: 由极值点定义知错误.

$$6. \text{ 确定常数 } A, B, C, \text{ 使得当 } x \rightarrow 0 \text{ 时有 } \frac{e^{\sin x}}{\sin x} = \frac{1 + Bx + Cx^2}{x + Ax^2} + o(x^2).$$

解: 两边乘以 $\sin x(x + Ax^2) \Rightarrow e^{\sin x}(x + Ax^2) = \sin x(1 + Bx + Cx^2) + o(x^4)$

$$\Rightarrow \left(1 + \sin x + \frac{\sin^2 x}{2} + \frac{\sin^3 x}{6}\right)(x + Ax^2) = \left(x - \frac{1}{6}x^3\right)(1 + Bx + Cx^2) + o(x^4)$$

$$\Rightarrow \left(1 + x - \frac{x^3}{6} + \frac{x^2}{2} + \frac{x^3}{6}\right)(x + Ax^2) = \left(x - \frac{1}{6}x^3\right)(1 + Bx + Cx^2) + o(x^4)$$

$$\Rightarrow \begin{cases} 1=1 \\ A+1=B \\ A+\frac{1}{2}=C+\frac{1}{6} \\ \frac{A}{2}=-\frac{B}{6} \end{cases} \Rightarrow \begin{cases} A=-\frac{1}{4} \\ B=\frac{3}{4} \\ C=\frac{5}{12} \end{cases}$$

$$7. \text{ 设 } z=z(x,y) \text{ 由 } x^2 + xy + z - 1 = \int_{x+1}^{2x+\sin y+z} e^{-(t-x)^2} dt \text{ 确定, 则 } dz|_{(0,1)} = \underline{\hspace{2cm}}.$$

$$\text{解: 令 } t-x=u \Rightarrow x^2 + xy + z - 1 = \int_1^{x+\sin y+z} e^{-u^2} du,$$

$$\text{对 } x \text{ 求偏导 } 2x + y + \frac{\partial z}{\partial x} = e^{-(x+\sin y+z)^2} \left[1 + (\cos y)y + \frac{\partial z}{\partial x} \right];$$

$$\text{对 } y \text{ 求偏导 } x + \frac{\partial z}{\partial y} = e^{-(x+\sin y+z)^2} \left[0 + (\cos y)x + \frac{\partial z}{\partial y} \right],$$

$$\begin{aligned} \text{令 } x=0, y=0 \Rightarrow z=1 \Rightarrow & \begin{cases} 1 + \frac{\partial z}{\partial x} = e^{-1} \left(1 + 1 + \frac{\partial z}{\partial x} \right) \\ 0 + \frac{\partial z}{\partial y} \Big|_{(0,1)} = e^{-1} \left(\frac{\partial z}{\partial y} \Big|_{(0,1)} \right) \Rightarrow \frac{\partial z}{\partial y} \Big|_{(0,1)} = 0 \end{cases} \\ \Rightarrow e + eA = 2 + A \Rightarrow A = & \frac{2-e}{e-1} = \frac{\partial z}{\partial x} \Big|_{(0,1)} \Rightarrow dz \Big|_{(0,1)} = \frac{2-e}{e-1} dx + 0 dy \end{aligned}$$

8. 设 $x_1 \in (0, 1)$, $e^{x_n}(x_n - x_{n+1}) = e^{x_n} - 1 (n=1, 2, \dots)$, 证明 $\{x_n\}$ 收敛并求 $\lim_{n \rightarrow \infty} x_n$.

解: (1) 有界性: $x_1 > 0$; 假设 $x_n > 0$; $x_{n+1} = e^{-x_n} - 1 + x_n (x_n > 0)$,

令 $f(x) = e^{-x} - 1 + x, f(0) = 1 - 1 + 0 = 0, f'(x) = -e^{-x} + 1 > 0$ 在 $x > 0$ 时成立,

$\Rightarrow x > 0$ 时, $f(x)$ 递增 $\Rightarrow x > 0$ 时, $f(x) > f(0) \Rightarrow x_{n+1} > 0$, 由数学归纳法得 $x_n > 0$ 恒成立.

(2) 单调性: $x_{n+1} - x_n = e^{-x_n} - 1 < 0 \Rightarrow \{x_n\}$ 单调递减, 由单调有界知 $\{x_n\}$ 收敛,

设 $\lim_{n \rightarrow \infty} x_n = A \Rightarrow e^A(A - A) = e^A - 1 \Rightarrow A = 0 \Rightarrow \lim_{n \rightarrow \infty} x_n = 0$

9. 设 $f(x)$ 在 $[0, 2]$ 上有连续导数, $f(1) = 0$, $M = \max_{0 \leq x \leq 2} |f'''(x)|$, 证明: $\left| \int_0^2 f(x) dx \right| \leq \frac{M}{3}$.

解: 令 $F(x) = \int_1^x f(t) dt \Rightarrow F(1) = F'(1) = 0, \int_0^2 f(x) dx = F(2) - F(0)$,

$$\begin{cases} F(2) = F(1) + F'(1) \cdot 1 + \frac{F''(1)}{2!} \cdot 1^2 + \frac{F'''(\xi_1)}{6} \cdot 1^3 \\ F(0) = F(1) + F'(1) \cdot (-1) + \frac{F''(1)}{2!} \cdot (-1)^2 + \frac{F'''(\xi_2)}{6} \cdot (-1)^3 \end{cases}, 0 < \xi_2 < 1 < \xi_1 < 2$$

$$\text{两式相减得 } |F(2) - F(0)| = \left| \frac{f''(\xi_1) + f''(\xi_2)}{6} \right| \Rightarrow \left| \int_0^2 f(x) dx \right| \leq \frac{1}{6} \cdot (M + M) = \frac{M}{3}$$

10. 设 $f(x)$ 在 $[0, 1]$ 上二阶连续可导, $|f''(x)| \leq M$, $f'(0) = f'(1) = 0$.

证明: (1) $|f'(x)| \leq \frac{M}{2}$; (2) $|f(0) - f(1)| \leq \frac{M}{4}$.

解: (1) $0 < x < \frac{1}{2}$ 时, $|f'(x) - f'(0)| = |(x-0) \cdot f''(\xi_1)| = x \cdot |f''(\xi_1)| \leq \frac{1}{2} M$

$\frac{1}{2} \leq x < 1$ 时, $|f'(x) - f'(1)| = |(x-1) \cdot f''(\xi_2)| = (1-x) \cdot |f''(\xi_2)| \leq \frac{1}{2} M$

$\Rightarrow$ 总有 $|f'(x)| \leq \frac{M}{2}$

$$(2) \begin{cases} f\left(\frac{1}{2}\right) = f(0) + f'(0) \cdot \frac{1}{2} + \frac{f''(\theta_1)}{2!} \cdot \frac{1}{4} \\ f\left(\frac{1}{2}\right) = f(1) + f'(1) \cdot \left(-\frac{1}{2}\right) + \frac{f''(\theta_2)}{2!} \cdot \frac{1}{4} \end{cases}, 0 < \theta_1 < \frac{1}{2} < \theta_2 < 1$$

$$\Rightarrow 0 = f(1) - f(0) + \frac{f''(\theta_2) - f''(\theta_1)}{8} \Rightarrow |f(1) - f(0)| = \frac{|f''(\theta_2) - f''(\theta_1)|}{8} \leq \frac{M + M}{8} = \frac{M}{4}$$

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